

**Table 3:** The 30 Boruta genes, ordered by SHAP rank. *Type and main role:* the UniProt protein name and its function, summarised in one sentence (The UniProt Consortium, 2024); bold names are pathway terms that pass Benjamini-Hochberg correction against the measured background. *This study:* four measurements, each bar scaled to the largest value among the 30 genes. *effect* is Hedges’  $g$  between 33 PD and 30 control donors; *folds* is the share of training folds in which Boruta reselected the gene; *bulk* is the inverse-variance pooled  $g$  in eight external bulk cohorts after dopamine-neuron content is regressed out (grey bar: sign opposite to discovery); *kept* is  $g$  after each donor’s CALB1/SOX6 composition is regressed out of every gene, the bar showing the share of the original effect that remains. Each  $q$  is a Benjamini-Hochberg value across the 30 genes within its own family of tests: the Wilcoxon rank-sum test for the discovery effect, and  $z$  tests on the pooled or adjusted effect against its standard error for the other two. *DA-neuron subtype:* CALB1 minus SOX6 lineage score and the subtype in which the gene is the strongest marker, from single-nucleus profiling of human nigra (Kamath et al., 2022). *Open Targets:* the overall association score with Parkinson disease, and beneath it the genetic-association component, which integrates GWAS and rare-variant evidence, where it is non-zero (Buniello et al., 2024). *Druggability:* Open Targets tractability evidence by modality and the number of drug or clinical candidates; for each modality only the strongest tier in Open Targets’ own hierarchy is shown. *Published link:* primary studies tying the gene to Parkinson’s disease or dopamine-neuron biology, study type first, and any measurement of the gene in patients’ blood or cerebrospinal fluid. They were found with one protocol for every gene (Methods): Europe PMC (Rosonovski et al., 2023), searched for the gene symbol, its protein name and every HGNC alias together with Parkinson’s-disease and dopamine-neuron terms, in titles, abstracts and open-access full text; each hit was read and kept only when it reports a finding about the gene itself, and reviews, acronym collisions and retracted papers were excluded. The whole search, every hit and whether it was kept, is Supplementary Table S1. Empty cells mean the resource holds no entry. The literature search ran on 2026-09-21; Open Targets release 26.06 was accessed on 2026-09-21;  $\uparrow$  higher and  $\downarrow$  lower in Parkinson’s disease; loci are GRCh38.

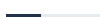
| Gene                                              | Type and main role                                                                                                                                                                                                         | This study                                                                                                                                                                                                                                                                                                                                                                                                                                            | DA-neuron subtype      | Open Targets              | Druggability                                                                                               | Published link to Parkinson’s disease or dopamine neurons                                                                                                                                                                                                                                                                        |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>RBM4B</i> $\uparrow$<br>11q13.2<br>SHAP rank 1 | <b>RNA-binding protein 4B</b><br>RNA-binding protein that<br>activates translation of the clock<br>gene PER1. (The UniProt<br>Consortium, 2024) [Q9BQ04]                                                                   | effect  $g +1.03, q 0.003$<br>folds  100% of folds<br>bulk  $g +0.66, q 0.004$<br>kept  $g +0.42, q 0.154$        | -1.8<br>SOX6 DDT       | 0.000                     |                                                                                                            |                                                                                                                                                                                                                                                                                                                                  |
| <i>PLXNC1</i> $\uparrow$<br>12q22<br>SHAP rank 2  | <b>Plexin-C1</b><br>Receptor for semaphorin 7A; its<br>signalling rearranges the<br>cytoskeleton. (The UniProt<br>Consortium, 2024) [O60486]<br><b>excitatory synapse,<br/>climbing-fibre synapse,<br/>synapse pruning</b> | effect  $g +1.23, q 0.003$<br>folds  96% of folds<br>bulk  $g +0.56, q 0.015$<br>kept  $g +0.48, q 0.107$         | +7.7<br>CALB1<br>CALCR | 0.001                     | <i>small molecule:</i><br>high-quality pocket<br><i>antibody:</i><br>surface/secreted (high<br>confidence) | Mouse development: semaphorin<br>7A–PLXNC1 signalling separates the<br>nigrostriatal from the mesolimbic<br>dopamine pathway (Chabrat et al.,<br>2017)                                                                                                                                                                           |
| <i>GPNMB</i> $\uparrow$<br>7p15.3<br>SHAP rank 3  | <b>Transmembrane<br/>glycoprotein NMB</b><br>Transmembrane glycoprotein,<br>annotated as a possible<br>melanogenic enzyme. (The<br>UniProt Consortium, 2024)<br>[Q14956]                                                   | effect  $g +0.97, q 0.003$<br>folds  92% of folds<br>bulk  $g +0.43, q 0.070$<br>kept  $g +0.53, q 0.086$ | -0.5<br>SOX6<br>AGTR1  | 0.503<br>genetic<br>0.782 | <i>antibody:</i> advanced<br>clinical<br><b>1 drug or clinical<br/>candidate</b>                           | Human GWAS: risk locus, the risk<br>allele raising GPNMB; cell and<br>mouse models: GPNMB binds<br>$\alpha$ -synuclein and promotes its<br>neuronal uptake (Diaz-Ortiz et al.,<br>2022)<br>Plasma: raised in Parkinson’s disease<br>across 731 patient and 59 control<br>samples, tracking severity<br>(Diaz-Ortiz et al., 2022) |

Table 3, continued

| Gene                                          | Type and main role                                                                                                                                                                       | This study                                                                                                                                                                                                                                                                                                                                                                                                                                            | DA-neuron subtype        | Open Targets | Druggability                                                                              | Published link to Parkinson's disease or dopamine neurons                                                                                                                                                                                                                                                                         |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>SCAI</i> ↓<br>9q33.3<br>SHAP rank 4        | Protein SCAI<br>Suppressor of MRTF-A-driven SRF transcription that regulates cell migration. (The UniProt Consortium, 2024) [Q8N9R8]                                                     | effect  $g -0.66, q 0.003$<br>folds  72% of folds<br>bulk  $g +0.13, q 0.699$<br>kept  $g -0.61, q 0.061$         | -0.2<br>CALB1<br>PPP1R17 | 0.000        | <i>antibody</i> : signal peptide or transmembrane                                         |                                                                                                                                                                                                                                                                                                                                   |
| <i>ICE1</i> ↑<br>5p15.32<br>SHAP rank 5       | Little elongation complex subunit 1<br>Scaffold of the little elongation complex, which drives small nuclear RNA transcription. (The UniProt Consortium, 2024) [Q9Y2F5]                  | effect  $g +0.73, q 0.017$<br>folds  80% of folds<br>bulk  $g +0.14, q 0.699$<br>kept  $g +0.69, q 0.053$         | +0.0                     | 0.000        |                                                                                           |                                                                                                                                                                                                                                                                                                                                   |
| <i>ZMYND8</i> ↑<br>20q13.12<br>SHAP rank 6    | MYND-type zinc finger-containing chromatin reader ZMYND8<br>Chromatin reader that recognises combined histone methylation and acetylation marks. (The UniProt Consortium, 2024) [Q9ULU4] | effect  $g +0.89, q 0.003$<br>folds  84% of folds<br>bulk  $g +0.29, q 0.258$<br>kept  $g +0.66, q 0.053$         | -1.6<br>SOX6<br>GFRA2    | 0.004        |                                                                                           |                                                                                                                                                                                                                                                                                                                                   |
| <i>CNTN4</i> ↓<br>3p26.3-p26.2<br>SHAP rank 7 | Contactin-4<br>Cell-surface adhesion molecule of nervous-system development that promotes neurite outgrowth. (The UniProt Consortium, 2024) [Q8I WV2]                                    | effect  $g -0.63, q 0.005$<br>folds  72% of folds<br>bulk  $g +0.10, q 0.752$<br>kept  $g -0.16, q 0.539$         | -3.1<br>SOX6<br>AGTR1    | 0.011        | <i>antibody</i> : surface/secreted (high confidence)                                      | Patient iPSC dopamine neurons, two brothers with the same PRKN mutations: CNTN4 was among the most down-regulated genes in the more severely affected brother (Cukier et al., 2022)<br>Cerebrospinal fluid: measured across dementias, including Parkinson's disease dementia and dementia with Lewy bodies (Muqaku et al., 2024) |
| <i>CFLAR</i> ↑<br>2q33.1<br>SHAP rank 8       | CASP8 and FADD-like apoptosis regulator<br>Apoptosis regulator that inhibits death-receptor-mediated cell death. (The UniProt Consortium, 2024) [O15519]                                 | effect  $g +0.54, q 0.009$<br>folds  56% of folds<br>bulk  $g -0.11, q 0.752$<br>kept  $g +0.32, q 0.225$ | +0.7<br>CALB1 GEM        | 0.005        | <i>small molecule</i> : structure with ligand<br><i>antibody</i> : surface/secreted (HPA) |                                                                                                                                                                                                                                                                                                                                   |
| <i>PPFIA4</i> ↑<br>1q32.1<br>SHAP rank 9      | Liprin-alpha-4<br>Scaffold that may regulate focal-adhesion turnover and position receptor tyrosine phosphatases. (The UniProt Consortium, 2024) [O75335]<br><b>excitatory synapse</b>   | effect  $g +0.82, q 0.003$<br>folds  76% of folds<br>bulk  $g +0.46, q 0.057$<br>kept  $g +0.46, q 0.122$ | +1.0<br>CALB1<br>PPP1R17 | 0.007        | <i>antibody</i> : surface/secreted (medium confidence)                                    |                                                                                                                                                                                                                                                                                                                                   |

Table 3, continued

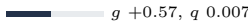
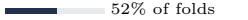
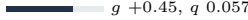
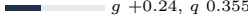
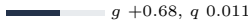
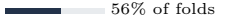
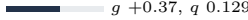
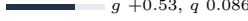
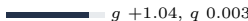
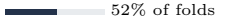
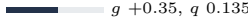
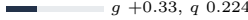
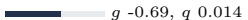
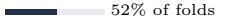
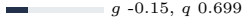
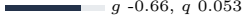
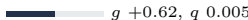
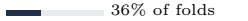
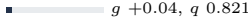
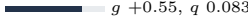
| Gene                                        | Type and main role                                                                                                                                                                                                   | This study                                                                                                                                                                                                                                                                                                                                                                                                                                                | DA-neuron subtype       | Open Targets | Druggability                                           | Published link to Parkinson's disease or dopamine neurons                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|--------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>CALB1</i> ↑<br>8q21.3<br>SHAP rank 10    | Calbindin<br>Calcium-binding protein that buffers cytosolic calcium. (The UniProt Consortium, 2024) [P05937]<br><b>presynaptic Ca<sup>2+</sup> binding</b>                                                           | effect  $g +0.57, q 0.007$<br>folds  52% of folds<br>bulk  $g +0.45, q 0.057$<br>kept  $g +0.24, q 0.355$         | +15.4<br>CALB1<br>CALCR | 0.084        |                                                        | Human post-mortem nigra: calbindin-rich compartments lose fewer dopamine neurons (Damier et al., 1999)                                                                                                                                                                                                                                                                                                                                              |
| <i>DNAJB1</i> ↑<br>19p13.12<br>SHAP rank 11 | DnaJ homolog subfamily B member 1<br>HSP40 co-chaperone that stimulates HSP70 in protein folding. (The UniProt Consortium, 2024) [P25685]                                                                            | effect  $g +0.68, q 0.011$<br>folds  56% of folds<br>bulk  $g +0.37, q 0.129$<br>kept  $g +0.53, q 0.086$         | -3.0<br>SOX6 DDT        | 0.089        | <i>small molecule</i> : structure with ligand          | Cell model: DNAJB1 clears $\alpha$ -synuclein through the Hsp70 system, and its phosphorylation by EGFR suppresses aggregation (Huang et al., 2025)<br>In vitro: DNAJB1 binds $\alpha$ -synuclein fibrils and recruits Hsc70 to take them apart (Monistrol et al., 2025)                                                                                                                                                                            |
| <i>CALB2</i> ↑<br>16q22.2<br>SHAP rank 12   | Calretinin<br>Calcium-binding protein that buffers intracellular calcium in specific neuronal populations. (The UniProt Consortium, 2024) [P22676]<br><b>presynaptic Ca<sup>2+</sup> binding, excitatory synapse</b> | effect  $g +1.04, q 0.003$<br>folds  52% of folds<br>bulk  $g +0.35, q 0.135$<br>kept  $g +0.33, q 0.224$         | +3.7<br>CALB1 GEM       | 0.038        |                                                        | Human post-mortem midbrain, 4 patients and 3 controls: calretinin-positive dopamine neurons of the substantia nigra were selectively preserved (Mouatt-Prigent et al., 1994)<br>Rat culture: calretinin-containing dopamine neurons were spared from levodopa toxicity (Isaacs et al., 1997)<br>Plasma: among the proteins most reduced in Parkinson's disease compared with controls, by nucleic acid-linked immunoassay (Vijaratnam et al., 2026) |
| <i>SH3GL3</i> ↓<br>15q25.2<br>SHAP rank 13  | Endophilin-A3<br>Endophilin that binds curved membranes and takes part in endocytosis. (The UniProt Consortium, 2024) [Q99963]                                                                                       | effect  $g -0.69, q 0.014$<br>folds  52% of folds<br>bulk  $g -0.15, q 0.699$<br>kept  $g -0.66, q 0.053$ | -4.0<br>SOX6<br>AGTR1   | 0.002        | <i>antibody</i> : surface/secreted (medium confidence) |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <i>GSTK1</i> ↑<br>7q34<br>SHAP rank 14      | Glutathione S-transferase kappa 1<br>Glutathione S-transferase that conjugates glutathione to foreign and endogenous compounds. (The UniProt Consortium, 2024) [Q9Y2Q3]                                              | effect  $g +0.62, q 0.005$<br>folds  36% of folds<br>bulk  $g +0.04, q 0.821$<br>kept  $g +0.55, q 0.083$ | +0.5<br>SOX6 DDT        | 0.003        | <i>small molecule</i> : structure with ligand          |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

Table 3, continued

| Gene                                         | Type and main role                                                                                                                                                                                                                                | This study                                                                                                                                                                                                                                                                                                                                                                                                                                            | DA-neuron subtype     | Open Targets | Druggability                                                                                          | Published link to Parkinson's disease or dopamine neurons |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <i>ADGRB3</i> ↓<br>6q12-q13<br>SHAP rank 15  | Adhesion G protein-coupled receptor B3<br>Adhesion G protein-coupled receptor that regulates synapse and dendritic-spine formation. (The UniProt Consortium, 2024) [O60242]<br><b>excitatory synapse, climbing-fibre synapse, synapse pruning</b> | effect  $g -0.52, q 0.011$<br>folds  36% of folds<br>bulk  $g +0.00, q 0.993$<br>kept  $g -0.36, q 0.191$         | +0.0                  | 0.000        | <i>small molecule</i> : structure with ligand<br><i>antibody</i> : surface/secreted (high confidence) |                                                           |
| <i>L3MBTL1</i> ↑<br>20q13.12<br>SHAP rank 16 | Lethal(3)malignant brain tumor-like protein 1<br>Polycomb-group reader of methylated lysines that keeps genes repressed. (The UniProt Consortium, 2024) [Q9Y468]                                                                                  | effect  $g +0.44, q 0.077$<br>folds  16% of folds<br>bulk  $g +0.10, q 0.752$<br>kept  $g +0.50, q 0.099$         | +0.0<br>CALB1 GEM     | 0.000        | <i>small molecule</i> : structure with ligand                                                         |                                                           |
| <i>C11orf54</i> ↓<br>nan<br>SHAP rank 17     | Beta-keto L-gulonate decarboxylase<br>Decarboxylase of the pentose pathway that also regulates HIF1A degradation. (The UniProt Consortium, 2024) [Q9H0W9]                                                                                         | effect  $g -0.71, q 0.007$<br>folds  40% of folds<br>bulk  $g -0.15, q 0.725$<br>kept  $g -0.68, q 0.053$         | -0.7<br>SOX6<br>GFRA2 | 0.000        |                                                                                                       |                                                           |
| <i>IBTK</i> ↓<br>6q14.1<br>SHAP rank 18      | Inhibitor of Bruton tyrosine kinase<br>Inhibitor of Bruton tyrosine kinase in B-cell development. (The UniProt Consortium, 2024) [Q9P2D0]                                                                                                         | effect  $g -0.90, q 0.003$<br>folds  60% of folds<br>bulk  $g -0.13, q 0.699$<br>kept  $g -0.76, q 0.053$        | -1.4<br>SOX6<br>AGTR1 | 0.000        | <i>antibody</i> : surface/secreted (medium confidence)                                                |                                                           |
| <i>CPNE3</i> ↓<br>8q21.3<br>SHAP rank 19     | Copine-3<br>Calcium-dependent phospholipid-binding protein involved in cell migration. (The UniProt Consortium, 2024) [O75131]                                                                                                                    | effect  $g -0.56, q 0.030$<br>folds  40% of folds<br>bulk  $g +0.23, q 0.461$<br>kept  $g -0.36, q 0.191$ | -8.9<br>SOX6<br>AGTR1 | 0.000        | <i>antibody</i> : surface/secreted (high confidence)                                                  |                                                           |
| <i>FUS</i> ↑<br>16p11.2<br>SHAP rank 20      | RNA-binding protein FUS<br>DNA/RNA-binding protein acting in transcription, splicing, RNA transport and DNA repair. (The UniProt Consortium, 2024) [P35637]                                                                                       | effect  $g +0.89, q 0.005$<br>folds  36% of folds<br>bulk  $g +0.07, q 0.752$<br>kept  $g +0.84, q 0.042$ | +0.0                  | 0.031        | <i>small molecule</i> : structure with ligand                                                         |                                                           |

Table 3, continued

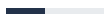
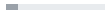
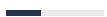
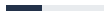
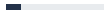
| Gene                                        | Type and main role                                                                                                                                                                          | This study                                                                                                                                                                                                                                                                                                                                                                                                                                            | DA-neuron subtype      | Open Targets              | Druggability                                                                                          | Published link to Parkinson's disease or dopamine neurons                                                                                                                                                                                                                |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|---------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>ELOVL7</i> ↓<br>5q12.1<br>SHAP rank 21   | Very long chain fatty acid elongase 7<br>Rate-limiting elongase of long- and very-long-chain fatty acids in the endoplasmic reticulum. (The UniProt Consortium, 2024) [A1L3X0]              | effect  $g -0.68, q 0.007$<br>folds  40% of folds<br>bulk  $g -0.29, q 0.461$<br>kept  $g -0.52, q 0.088$         | -3.1<br>SOX6<br>AGTR1  | 0.316<br>genetic<br>0.507 | <i>small molecule</i> : structure with ligand<br><i>antibody</i> : signal peptide or transmembrane    | Human case-control study: variant associated with Parkinson's disease (Li et al., 2018)                                                                                                                                                                                  |
| <i>PRKAR2A</i> ↑<br>3p21.31<br>SHAP rank 22 | cAMP-dependent protein kinase type II-alpha regulatory subunit<br>Regulatory subunit of cAMP-dependent protein kinase that anchors it to membranes. (The UniProt Consortium, 2024) [P13861] | effect  $g +0.69, q 0.021$<br>folds  32% of folds<br>bulk  $g -0.08, q 0.752$<br>kept  $g +0.63, q 0.053$         | -0.0<br>SOX6<br>GFRA2  | 0.020                     | <i>small molecule</i> : structure with ligand<br><i>antibody</i> : surface/secreted (high confidence) | Human Lewy body disease brain and mice: $\alpha$ -synuclein sequesters DNMT1 from the nucleus, hypomethylating CpG islands upstream of PRKAR2A (Desplats et al., 2011)                                                                                                   |
| <i>USP12</i> ↓<br>13q12.13<br>SHAP rank 23  | Ubiquitin carboxyl-terminal hydrolase 12<br>Deubiquitinating enzyme that regulates immune and inflammatory signalling. (The UniProt Consortium, 2024) [O75317]                              | effect  $g -1.06, q 0.003$<br>folds  48% of folds<br>bulk  $g -0.21, q 0.480$<br>kept  $g -0.60, q 0.061$         | -4.9<br>SOX6<br>AGTR1  | 0.023                     | <i>small molecule</i> : structure with ligand<br><i>antibody</i> : surface/secreted (high confidence) |                                                                                                                                                                                                                                                                          |
| <i>SOX6</i> ↓<br>11p15.2<br>SHAP rank 24    | Transcription factor SOX-6<br>Transcription factor in neurogenesis and cartilage development. (The UniProt Consortium, 2024) [P35712]                                                       | effect  $g -0.77, q 0.004$<br>folds  36% of folds<br>bulk  $g -0.41, q 0.210$<br>kept  $g -0.33, q 0.224$         | -18.3<br>SOX6<br>AGTR1 | 0.087                     |                                                                                                       | Human single-nucleus nigra: marks the dopamine population depleted in Parkinson's disease (Kamath et al., 2022)<br>Mouse: legumain cleaves SOX6 in vulnerable dopamine neurons, and the fragment speeds degeneration in A53T $\alpha$ -synuclein mice (Nie et al., 2024) |
| <i>MANF</i> ↑<br>3p21.2<br>SHAP rank 25     | Mesencephalic astrocyte-derived neurotrophic factor<br>Secreted factor that selectively promotes survival of ventral midbrain dopamine neurons. (The UniProt Consortium, 2024) [P55145]     | effect  $g +0.55, q 0.020$<br>folds  16% of folds<br>bulk  $g +0.39, q 0.103$<br>kept  $g +0.37, q 0.191$ | -1.0<br>SOX6 DDT       | 0.102                     | <i>antibody</i> : surface/secreted (high confidence)                                                  | Rat 6-hydroxydopamine model: MANF restores nigrostriatal function (Voutilainen et al., 2009)                                                                                                                                                                             |

Table 3, continued

| Gene                                         | Type and main role                                                                                                                                                                             | This study                                                                                                                                                                                                                                                                                                                                                                                                                                            | DA-neuron subtype      | Open Targets | Druggability                                                                                                                                  | Published link to Parkinson’s disease or dopamine neurons                                                                                                                                                                                                                  |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>LGMN</i> ↓<br>14q32.12<br>SHAP rank 26    | <b>Legumain</b><br>Lysosomal protease that cleaves proteins after asparagine residues. (The UniProt Consortium, 2024) [Q99538]                                                                 | effect  $g -0.46, q 0.031$<br>folds  20% of folds<br>bulk  $g +0.13, q 0.699$<br>kept  $g -0.24, q 0.355$         | +0.0                   | 0.082        | <i>small molecule</i> : structure with ligand<br><i>antibody</i> : surface/secreted (high confidence)                                         | Cell, mouse and human brain: legumain cleaves $\alpha$ -synuclein at N103, and the fragment aggregates and is toxic to dopamine neurons (Zhang et al., 2017)<br>Mouse: it cleaves $\alpha$ -synuclein and tau and drives their spread from gut to brain (Ahn et al., 2019) |
| <i>SSTR1</i> ↓<br>14q21.1<br>SHAP rank 27    | <b>Somatostatin receptor type 1</b><br>Somatostatin receptor that inhibits adenylyl cyclase through G proteins. (The UniProt Consortium, 2024) [P30872]                                        | effect  $g -0.93, q 0.003$<br>folds  48% of folds<br>bulk  $g -0.07, q 0.752$<br>kept  $g -0.44, q 0.136$         | -6.8<br>SOX6 DDT       | 0.075        | <i>small molecule</i> : structure with ligand<br><i>antibody</i> : surface/secreted (high confidence)<br><b>8 drug or clinical candidates</b> |                                                                                                                                                                                                                                                                            |
| <i>BCL11A</i> ↑<br>2p16.1<br>SHAP rank 28    | <b>BCL11 transcription factor A</b><br>Transcription factor associated with the BAF chromatin-remodelling complex. (The UniProt Consortium, 2024) [Q9H165]                                     | effect  $g +0.70, q 0.003$<br>folds  40% of folds<br>bulk  $g +0.16, q 0.699$<br>kept  $g +0.40, q 0.161$         | +0.7<br>SOX6<br>GFRA2  | 0.008        | <b>1 drug or clinical candidate</b>                                                                                                           | Mouse: marks a subset of nigral dopamine neurons that is especially vulnerable to $\alpha$ -synuclein overexpression and oxidative stress (Tolve et al., 2021)                                                                                                             |
| <i>CACNA1A</i> ↑<br>19p13.13<br>SHAP rank 29 | <b>Voltage-dependent P/Q-type calcium channel subunit alpha-1A</b><br>Pore-forming subunit of P/Q-type voltage-gated calcium channels. (The UniProt Consortium, 2024) [O00555]                 | effect  $g +0.48, q 0.068$<br>folds  24% of folds<br>bulk  $g +0.08, q 0.752$<br>kept  $g +0.55, q 0.083$         | -2.8<br>SOX6<br>AGTR1  | 0.355        | <i>small molecule</i> : approved drug<br><i>antibody</i> : surface/secreted (high confidence)<br><b>11 drug or clinical candidates</b>        | Cell model: the Parkinson’s kinase LRRK2 regulates the Ca <sub>v</sub> 2.1 channel (Bedford et al., 2016)                                                                                                                                                                  |
| <i>TRRAP</i> ↑<br>7q22.1<br>SHAP rank 30     | <b>Transformation/transcription domain-associated protein</b><br>Adapter of histone acetyltransferase complexes that mark chromatin for transcription. (The UniProt Consortium, 2024) [Q9Y4A5] | effect  $g +0.83, q 0.004$<br>folds  32% of folds<br>bulk  $g +0.06, q 0.772$<br>kept  $g +0.65, q 0.053$ | +0.6<br>CALB1<br>CALCR | 0.333        | <i>small molecule</i> : structure with ligand                                                                                                 |                                                                                                                                                                                                                                                                            |

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